for...in vs for...of

Why for...in prints 0, 1

for-in.js

```
const fruits = ["apple", "mango"];
for (const x in fruits) console.log(x); // "0" "1"
for (const x of fruits) console.log(x); // apple mango
```

output "0" "1" apple mango

Why it happens

01 for...in = keys

It loops over the keys (property names) of an object.

02 Array keys

An array's keys are its indexes, as strings: "0", "1".

03 for...of = values

It loops over the values of any iterable.

04 Rule

Arrays and strings: for...of. Plain objects: for...in.

Which loop when?

Loops compared

loop	gives	works on
`for...in`	keys (strings)	objects
`for...of`	values	arrays, strings, Map, Set
`forEach`	value, index	arrays (no break)
`Object.entries + for...of`	[key, value]	objects

Traps to mention

- **for...in includes inherited keys** added to the prototype. That is another reason to avoid it on arrays.
- **for...of on a plain object throws** TypeError: object is not iterable.
- **Keys are strings,** so "0" + 1 is "01", not 1.

Your 20-second answer

"for...in loops over keys, and for an array the keys are the indexes as strings, so it prints '0' and '1'. for...of loops over values, so it prints apple and mango. I use for...of for arrays and for...in or Object.entries for objects.